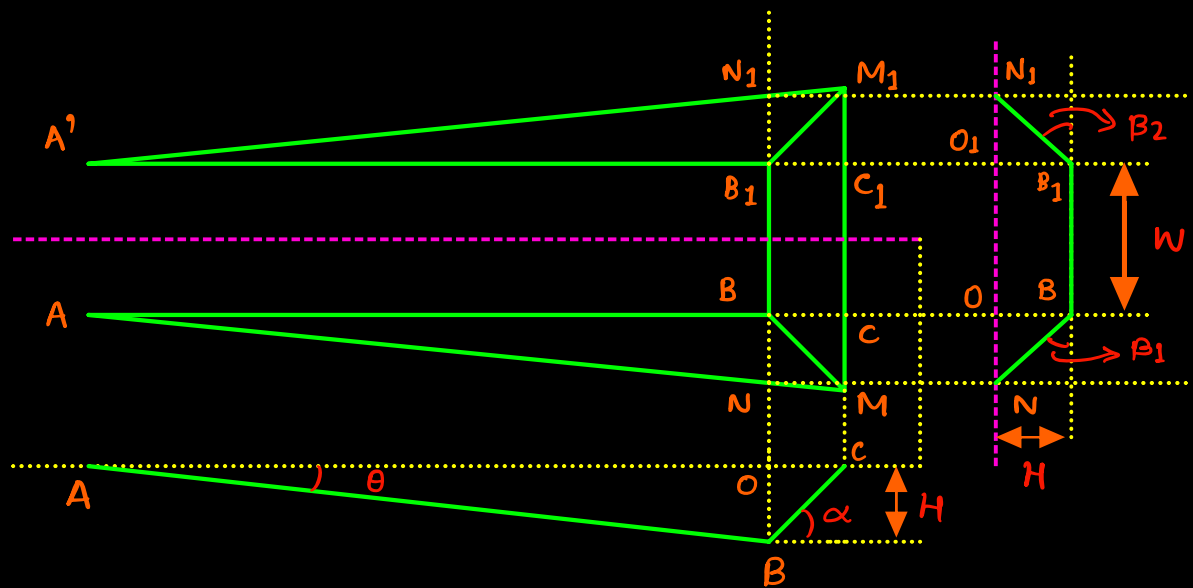
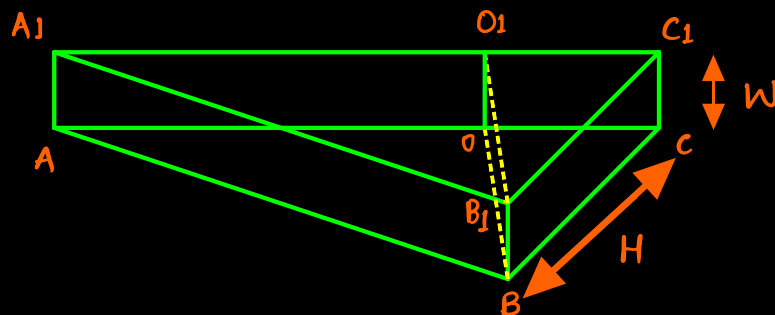


(Inverted Pyramid)



(Prism)

The points O & O_1 are projections of B & B_1 on the surface.

Purpose is to find AC : ———

Considering $\triangle AOB$;

$$\tan \theta = \frac{OB}{AO}$$

$$\Rightarrow AO = \frac{OB}{\tan \theta} = \frac{H}{\tan \theta}$$

Similarly, for $\triangle BOC$;

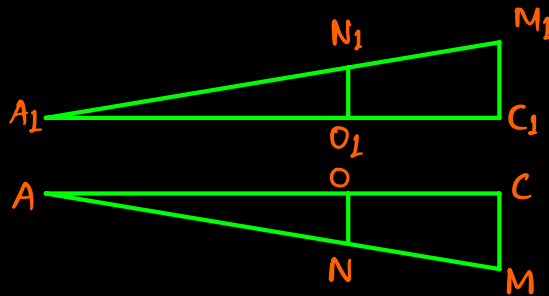
$$\tan \alpha = \frac{OB}{OC}$$

$$\Rightarrow OC = \frac{OB}{\tan \alpha} = \frac{H}{\tan \alpha}$$

$$AC = OA + OC = \frac{H}{\tan \theta} + \frac{H}{\tan \alpha}$$

$$AC = H \left[\frac{1}{\tan \theta} + \frac{1}{\tan \alpha} \right] = H \left[\frac{(\tan \alpha) + (\tan \theta)}{(\tan \alpha)(\tan \theta)} \right]$$

Considering the surface projection (plane surface) $AOCMNA$ and $A_1O_1C_1M_1N_1A_1$



From similar triangles, $\triangle AON$ & $\triangle ACM$;

$$\frac{MC}{ON} = \frac{AC}{AO}$$

$$\Rightarrow \frac{MC}{ON} = \frac{AO + OC}{AO} = \left(1 + \frac{OC}{AO} \right)$$

$$\Rightarrow MC = ON \left[1 + \frac{OC}{AO} \right]$$

Similarly,

$$\frac{M_1C_1}{O_1N_1} = \frac{A_1C_1}{A_1O_1}$$

$$\Rightarrow \frac{M_1C_1}{O_1N_1} = \frac{A_1O_1 + O_1C_1}{A_1O_1} = \left(1 + \frac{O_1C_1}{A_1O_1} \right)$$

$$\Rightarrow M_1C_1 = O_1N_1 \left[1 + \frac{O_1C_1}{A_1O_1} \right]$$

From the transverse section ;

Considering $\triangle OBN$;

$$\tan \beta_1 = \frac{OB}{ON}$$

$$\Rightarrow ON = \frac{OB}{\tan \beta_1} = \frac{H}{\tan \beta_1}$$

Considering $\triangle O_1 B_1 N_1$;

$$\tan \beta_2 = \frac{O_1 B_1}{O_1 N_1}$$

$$\Rightarrow O_1 N_1 = \frac{O_1 B_1}{\tan \beta_2} = \frac{H}{\tan \beta_2}$$

Putting ON & $O_1 N_1$ in MC & $M_1 C_1$;

$$MC = ON \left[1 + \frac{OC}{OA} \right]$$

$$MC = \frac{H}{\tan \beta_1} \left[1 + \frac{OC}{OA} \right]$$

$$MC = \frac{H}{\tan \beta_1} \left[1 + \frac{H}{\tan \alpha} \times \frac{\tan \theta}{H} \right]$$

$$MC = \frac{H}{\tan \beta_1} \left[1 + \frac{\tan \theta}{\tan \alpha} \right] = \frac{H [\tan \alpha + \tan \theta]}{(\tan \alpha) (\tan \beta_1)}$$

$$\begin{cases} OC = \frac{H}{\tan \alpha} \\ OA = \frac{H}{\tan \theta} \end{cases}$$

Similarly ;

$$M_1 C_1 = O_1 N_1 \left[1 + \frac{O_1 C_1}{A_1 O_1} \right]$$

$$M_1 C_1 = \frac{H}{\tan \beta_2} \left[1 + \frac{H}{\tan \alpha} \times \frac{\tan \theta}{H} \right]$$

$$M_1 C_1 = \frac{H [\tan \alpha + \tan \theta]}{(\tan \alpha) (\tan \beta_2)}$$

$$MM_1 = M_1 C_1 + MC$$

$$= \frac{H(\tan\alpha + \tan\theta)}{\tan\alpha} \left[\frac{1}{\tan\beta_1} + \frac{1}{\tan\beta_2} \right]$$

In general, β_1 & β_2 are equal = β

$$MM_1 = \frac{2 \cdot H \cdot (\tan\alpha + \tan\theta)}{(\tan\alpha)(\tan\beta)}$$

To calculate cut volume, whole volume is divided into three —

- (i) Segment $ABCC_1B_1A$ $\longrightarrow$ Inverted prism
 - (ii) Segment $ABCMN$
 - (iii) Segment $A_1B_1C_1M_1N_1$
- } $\longrightarrow$ Joint together to form inverted pyramid with triangular base AMM_1A .

volume of segment $ABCC_1B_1A = V_1$

$$V_1 = W \times \text{ar}(\triangle ABC)$$

$$= W \times \frac{1}{2} \times AC \times OB$$

$$= \frac{W}{2} \times \frac{H(\tan\alpha + \tan\theta)}{(\tan\alpha)(\tan\theta)} \times H$$

$$= \frac{H^2 W}{2} \left(\frac{\tan\alpha + \tan\theta}{(\tan\alpha)(\tan\theta)} \right)$$

volume of inverted pyramid ($ABCMN$) = V_2

$$V_2 = \frac{1}{3} \times H \times \text{ar}(\triangle AMM_1)$$

$$= \frac{1}{3} \times H \times \frac{1}{2} \times (MC + M_1C_1) \times AC$$

$$= \frac{1}{3} \times H \times \frac{1}{2} \times \left[\frac{H(\tan\alpha + \tan\theta)}{(\tan\alpha)(\tan\beta_1)} + \frac{H(\tan\alpha + \tan\theta)}{(\tan\alpha)(\tan\beta_1)} \right] \times AC$$

$$= \frac{H^2(\tan\alpha + \tan\theta)}{6 \tan\alpha} \left[\frac{1}{\tan\beta_1} + \frac{1}{\tan\beta_2} \right] \times \frac{H(\tan\alpha + \tan\theta)}{(\tan\alpha)(\tan\theta)}$$

$$\beta_1 = \beta_2 = \beta$$

$$V_2 = \frac{H^3}{3} \frac{(\tan \alpha + \tan \theta)^2}{(\tan \alpha)^2 (\tan \theta) (\tan \beta)}$$

$$\text{Total volume, } V = V_1 + V_2$$

$$= \frac{H^2 w (\tan \alpha + \tan \theta)}{2 (\tan \alpha) (\tan \theta)} + \frac{H^3}{3} \frac{(\tan \alpha + \tan \theta)^2}{(\tan \alpha)^2 (\tan \theta) (\tan \beta)}$$

$$= \frac{H^2 (\tan \alpha + \tan \theta)}{(\tan \alpha) (\tan \theta)} \left[\frac{w}{2} + \frac{H}{3} \frac{(\tan \alpha + \tan \theta)}{(\tan \alpha) (\tan \beta)} \right]$$